Zisen Dai

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Education

Tsinghua University — August 2023 – Present
Bachelor of Science in Physics | GPA: 4.0/4.0 | Rank: 1/28 — *Beijing, China*

UC Berkeley — January 2026 – May 2026
Exchange Student in Astrophysics (Adviser: Wenbin Lu) — *Berkeley, CA, USA*

Research Experience

Department of Astronomy, UC Berkeley — May 2026 – Present
Research Assistant (Advised by Wenbin Lu) — *Berkeley, CA, USA*

- Tracking cutting-edge developments in high-energy transients via weekly group discussions.

Institute for Advanced Study, Tsinghua University — March 2025 – Present
Research Assistant (Advised by Xuening Bai) — *Beijing, China*

- Investigating cosmic ray transport dynamics within protoplanetary disk (PPD) winds.
- Developed a Python simulation framework from scratch to model particle transport via the Guiding Center Approximation, analyzing the impact of scattering mechanisms and strengths.

Project

Rosseland-Mean Opacity Computation — April 2026
Calculate the Rosseland-Mean Opacity of Pure Hydrogen Gas under LTE (report, code) — *Berkeley, CA, USA*

- Developed a Python framework to computing the Rosseland-mean opacity of pure hydrogen gas in LTE, and benchmarking it against the LANL/TOPS opacity tables.
- Localized the main opacity discrepancy domains and analyzed the missing physical channels and omitted physics causing the issue.

LBGs Selection Based on Photometric Data — January 2026
Machine Learning-Based High-Redshift Target Selection (report, code) — *Beijing, China*

- Developed a LightGBM classifier on CLAUDS photometry, utilizing sample weighting and probability calibration to identify high-redshift ($z \sim 2 - 3$) Lyman Break Galaxies.
- Corrected selection bias via TSR/SSR metrics, achieving 0.995 AUC and 92.0% purity — outperforming the traditional u-dropout method with a $3.3\times$ sample size increase.

JUNO Optical Simulation & Signal Reconstruction — August 2025
Comprehensive Physics Simulation and Machine Learning Modeling (code) — *Beijing, China*

- Developed a Monte Carlo Python framework to simulate complex photon transport, Fresnel reflection, and emission for 17,612 PMTs.
- Engineered an ultra-high-capacity LightGBM model with multi-center Gaussian attention to reconstruct spatial-temporal Probe functions, reducing temporal integral error to $\sim$3%.

Spectrum and Light Curve Analysis of Supernovae — June 2025
Photometric and Spectroscopic Classification (report, code) — *Beijing, China*

- Classified SN 2025kid as a Type Ia supernova by analyzing multi-band photometric data and spectroscopic absorption features.
- Extracted key parameters (peak magnitude, decline rate Δm_{15}) of SN 2025fvw via light curve fitting, utilizing the Phillips relation for standardization.

Awards & Honors

National Scholarship *2024, 2025*
Highest Honor for Undergraduates, The Ministry of Education of China

Tsinghua Xuetang Talents Program *2023 – 2025*
Chi-Sun Yeh Class, Elite Physics Track

Excellent Academic Performance Scholarship *2024, 2025*
Tsinghua University

Basic Disciplines Interdisciplinary Challenge *2025*
Third Prize, Tsinghua University

Chinese Physics Olympiad (CPhO) *2022*
First Prize in Zhejiang

Selected Coursework

Perfect grades (4.00/4.00) across all 23 major courses (including **6 A+, 13 A**)

Astrophysics & Data: Statistical Methods in Astrophysics (A+), Observational Astronomy (A+), Big Data in Experimental Physics I & II (A+), Astrophysics

Advanced Physics: Statistical Mechanics (A+), Quantum Mechanics, Electrodynamics, Analytical Mechanics, Atomic and Molecular Physics

Advanced Mathematics: Mathematical Physics Equations (A+), Complex Analysis, Linear Algebra for Physical Sciences

Other Experience

Volunteer Peer Tutor March 2025 – Present
Drop-in Tutoring Center, Tsinghua University *Beijing, China*

- Providing one-on-one academic support and problem-solving strategies for undergraduates in advanced physics and math courses.

Student Life Coordinator March 2024 – December 2025
Department of Organization, Zhili College *Beijing, China*

- Organized college-wide student events, coordinated group activities, and facilitated team-building initiatives.

Academic Representative September 2023 – September 2025
Physics Class 32, Zhili College *Beijing, China*

- Liaised between faculty and students to facilitate academic communications and organize study groups.